

QUESTION BANK

1. Find $L^{-1} \left[\frac{3s-8}{4s^2+25} \right]$

Ans : $f(t) = \frac{3}{4} \cos \left(\frac{5}{2} \right) t - \frac{4}{5} \sin \left(\frac{5}{2} \right) t$

2. Find $L^{-1} \left[\frac{4s-18}{9-s^2} \right]$

Ans : $f(t) = 6 \sinh 3t - 4 \cosh 3t$

3. Find $L^{-1} \left[\frac{s^2-3s+4}{s^3} \right]$

Ans : $f(t) = 1 - 3t + 2t^2$

4. Find $L^{-1} \left[\frac{1}{s-2} + \frac{1}{s^2-9} + \frac{1}{s^2+25} + \frac{s}{s^2+9} \right]$

Ans : $f(t) = e^{2t} + \frac{1}{3} \sinh 3t + \frac{1}{5} \sin 5t + \cos 3t$

5. Find $L^{-1} \left[\frac{1}{s} e^{-\frac{1}{s}} \right]$ {Use the series expansion of e^x }

Ans : $f(t) = \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{(n!)^2}$

6. If $L^{-1} \left[\frac{s^2-1}{(s^2+1)^2} \right] = t \cos t$ then show that $L^{-1} \left[\frac{9s^2-1}{(9s^2+1)^2} \right] = \frac{t}{9} \cos \frac{t}{3}$

7. Find $L^{-1} \left[\frac{1-3s}{s^2+8s+21} \right]$

Ans : $f(t) = -3e^{-4t} \cos(\sqrt{5}t) + \frac{13}{\sqrt{5}} e^{-4t} \sin(\sqrt{5}t)$

8. Find $L^{-1} \left[\log \frac{s(s+1)}{s^2+4} \right]$

Ans : $f(t) = \frac{2 \cos 2t - e^{-t} - 1}{t}$

9. Find $L^{-1} [\cot^{-1} s]$

Ans : $f(t) = \frac{\sin t}{t}$

10. Find $L^{-1} \left[\frac{s}{s^2+6s+13} \right]$

Ans : $f(t) = e^{-3t} \left[\cos 2t - \frac{3}{2} \sin 2t \right]$

11. Find $L^{-1} \left[\frac{s}{s^4+s^2+1} \right]$

Ans : $f(t) = \frac{2}{\sqrt{3}} \sin \left(\frac{\sqrt{3}}{2} t \right) \sinh \left(\frac{t}{2} \right)$

12. Find $L^{-1} \left[\frac{s^2}{s^4+4a^4} \right]$

Ans : $f(t) = \frac{\sinh(at) \cos(at) + \cosh(at) \sin(at)}{2a}$

13. Find $L^{-1} \left[\log \sqrt{\frac{s^2+1}{s^2+4}} \right]$

Ans : $f(t) = \frac{\cos 2t - \cos t}{t}$

14. Find $L^{-1} \left[\frac{1}{s^3(s^2+a^2)} \right]$

Ans : $f(t) = \frac{1}{a^4} \left[\frac{a^2 t^2}{2} + \cos at - 1 \right]$

15. Find $L^{-1} \left[\frac{e^{-s}}{(s-1)(s-2)} \right]$

Ans: $f(t) = [e^{2(t-1)} - e^{(t-1)}]u(t-1)$

16. Find $L^{-1} \left[\frac{(1-e^{-s})(2-e^{-2s})}{s^2} \right]$

Ans : $f(t) = t^2 - (t-1)^2 u(t-1) - \frac{(t-2)^2 u(t-2)}{2} + \frac{(t-3)^2 u(t-3)}{2}$

17. Using Convolution Theorem find $L^{-1} \left[\frac{s+2}{(s^2+4s+5)^2} \right]$

Ans : $f(t) = \frac{e^{-2t} t \sin t}{2}$

18. Using Convolution Theorem find $L^{-1} \left[\frac{s^2}{(s^2+16)(s^2+9)} \right]$

Ans : $f(t) = \frac{4 \sin 4t - 3 \sin 3t}{7}$

19. Solve : $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = 0$ under the conditions $y(0) = 1$; $y'(0) = 1$.

Ans : $y = 2e^{-t} - e^{-2t}$

20. Solve : $y'' + 2y' + 5y = e^{-t} \sin t$ given $y(0) = 0$; $y'(0) = 0$

Ans : $y = \frac{e^{-t}(\sin t - \sin 2t)}{3}$
